

Product Requirements Document

Goals, personas, scope, functional & non-functional requirements, constraints and risks ·
UrbanLens Platform v1.0

1 · Product summary

Product: UrbanLens — an AI-powered mobile urban intelligence platform that retrofits public-transport fleets with edge-AI sensing. **Users:** municipal corporations/PWD (road maintenance), traffic police (congestion + enforcement), and citizens (safer roads). **Outcome:** a city-wide, continuously-updated map of road defects, congestion and traffic violations at near-zero marginal cost.

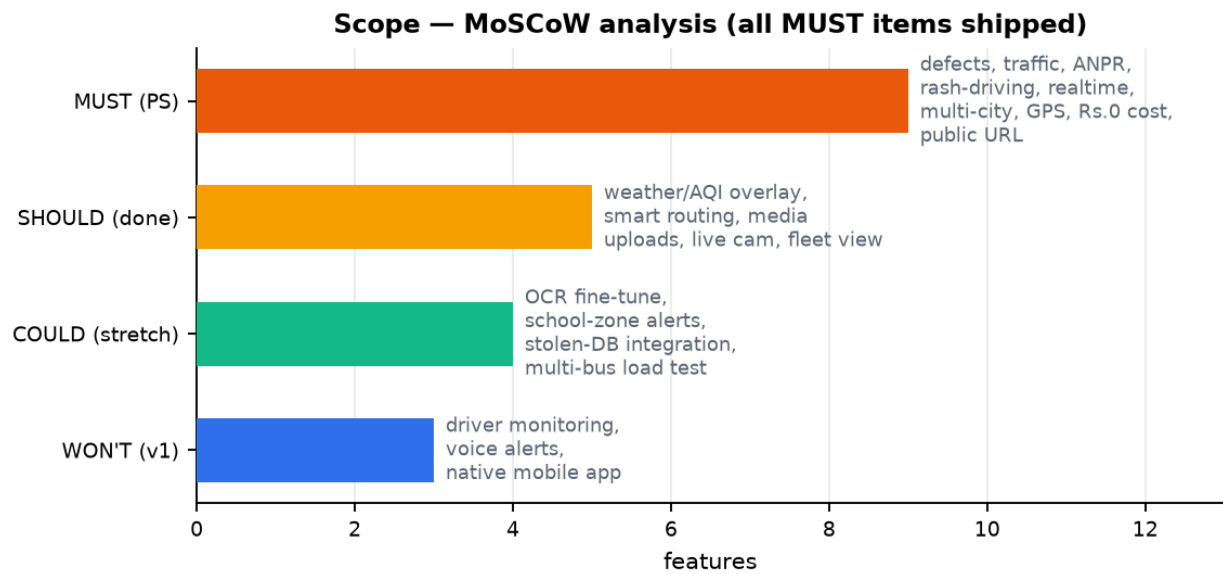
2 · Goals & success metrics

Goal	Metric	Result (prototype)
Detect road defects automatically	mAP50 on 9 classes	0.669 (v3), open_manhole 0.971
Measure live traffic	counting accuracy vs ground truth	19/19 vehicles on demo route
Read number plates (ANPR)	valid-plate read confidence	93–96%, 3-frame voting
Flag rash driving	false-positive rate (unit scenarios)	0/1 steady flagged, weaver+speeder caught
Real-time command view	detection → dashboard latency	1–3 s (target < 5 s)
Reliable ingest	burst upload success	40/40 POSTs after pooler fix
Zero running cost	external API spend	Rs.0 — all free tiers
Dashboard quality	automated UI checks	83/83 green

3 · Personas & user stories

- **Ward engineer (PWD):** opens the dashboard each morning, sees severity-ranked defects in her ward with deduplicated work orders, assigns and tracks to resolution with SLA.
 - **Traffic control-room operator:** watches live congestion heat + bottleneck corridors, receives ANPR/rash-driving alerts with plate, confidence, GPS and timestamp.
 - **Police investigator:** queries a plate or a stretch of road; gets evidence-linked events for hit-and-run or stolen-vehicle cases.
 - **Citizen:** benefits silently — hazards found and fixed before complaints, without anyone having to file them.
1. As a bus drives its route, every detected defect is geo-tagged, deduplicated and severity-scored — no human review needed for triage.
 2. As a vehicle is read by ANPR, its plate, confidence, position and time are recorded; watchlist hits alert instantly.
 3. As traffic builds, congestion indices per corridor update every 10 seconds per bus.

4 · Scope



5 · Functional requirements (FR)

ID	Requirement	Status
FR-1	Detect 9 road-defect classes onboard with confidence ≥ 0.45 , temporal streaks and cooldowns to suppress noise	■ shipped
FR-2	Geo-tag every event (GPS device or route replay); never tag without a fix	■ shipped
FR-3	Count vehicles & pedestrians in 10-second windows; compute congestion index (counts + speed deficit + crowd bump)	■ shipped
FR-4	ANPR: detect plate → OCR → normalise to Indian format → 3-frame majority vote → alert with plate+confidence+GPS+time	■ shipped
FR-5	Rash-driving: flag fleet-relative speeding and lane-weaving with evidence; attach plate when available	■ shipped
FR-6	Watchlist matching on plates with stolen-vehicle alert kind	■ shipped (demo list)
FR-7	Offline spool + retry; uploads never block detection	■ shipped
FR-8	Multi-city dashboard: live map, congestion, incidents, alerts, analytics, fleet view, media, live cam	■ shipped
FR-9	Incident workflow: new → confirmed → assigned → in_progress → resolved with audit trail	■ shipped
FR-10	Realtime push (websockets) with polling fallback	■ shipped

FR-11	Weather + AQI overlays from free government/open APIs	■ shipped
FR-12	Smart routing that avoids defect-dense and congested segments	■ shipped
FR-13	Publicly accessible demo URLs	■ shipped
FR-14	School-zone/vulnerable-pedestrian alerting	■ partial (persons counted live; zone logic pending)

6 · Non-functional requirements (NFR)

ID	Requirement	Verified
NFR-1	Latency: detection → dashboard < 5 s	■ 1–3 s
NFR-2	Reliability: ingest survives burst load and cold DB connections	■ 40/40 hammer test
NFR-3	Cost: Rs.0 external API spend; free-tier infrastructure	■ audited
NFR-4	Privacy: blur bystanders; ANPR retains plate text only (no faces)	■ implemented
NFR-5	Offline operation: detection continues without connectivity; spool drains on reconnect	■ tested
NFR-6	Portability: same code on laptop and Raspberry Pi 5	■ CPU-only ONNX
NFR-7	Security: token-gated ingest; browser uses anon key only	■ implemented
NFR-8	Explainability: every alert carries evidence + confidence	■ implemented

7 · Constraints

- Free-tier infra (zero-budget mandate for prototype phase): Supabase pooler limits, tunnel URLs rotate on restart.
- Edge hardware: laptop demo today; Pi 5 pilot planned — no GPU assumption.
- Data: publicly available datasets for training (RDD2022 + Kaggle Indian plates); no city-provided data yet.
- Legal: ANPR evidence positioned as intelligence for authorities, not automated enforcement (due-process sensitivity).

8 · Risks & mitigations

Risk	Impact	Mitigation
Detector degrades at night/rain	missed defects	quality gates suppress low-confidence frames; night-augmented retrain planned
OCR errors on dirty/angled plates	wrong plate in alert	3-frame voting + confidence floor + regex validation of Indian format

Free-tier DB limits at fleet scale	ingest stalls	tx-pooler tuning done; pilot can upgrade to cheap paid tier (~\$25/mo) when justified
Rash-driving false positives	officer distrust	evidence-based alerts (speeds, lane changes), cooldown per vehicle, human confirms before action
Privacy backlash	adoption barrier	on-device blur, no face retention, published data-retention policy